POOJA KIRAN | SECURITY ENGINEERING

A Unified Control Plane for Machine-Learning Security Services

An Authenticated Gateway with Configurable ML Security Service Endpoints

Source inventory: d3991135d20970a5c4c8a36fb49c5d254d282edd

Test snapshot: d54336b5f5f058a4d756bf2c7a023ad8510bbd6f

CI run 37173681100 | initiated 2026-10-04 UTC | Apache-2.0

https://github.com/poojakira/unified-ml-security-platform

82
tests passed in cited CI

95.96%
statement coverage at cited CI

4
synchronous service slots in configured topology

Stubs
local contract topology, not production products

CONTRIBUTION

- A real authenticated FastAPI control plane with four production HTTP service slots, a local contract-test topology built from stubs, and a production Compose contract that requires externally released product images instead of pretending the stubs are product implementations.

IMPLEMENTED METHOD

- 1. Authenticate external callers at the gateway.
- 2. Strip caller-controlled sensitive headers and inject per-service credentials.
- 3. Route four synchronous service prefixes over an internal-only network.
- 4. Use local health-contract stubs to test topology without misrepresenting product logic.
- 5. Require explicit external product images in production Compose.
- 6. Validate unit, security, integration, image-build, and deployment-contract behavior in CI.

LIMITATIONS AND EVIDENCE BOUNDARY

- Product business logic is owned by external repositories/images.
- No production deployment, uptime, or load SLO is established.
- Multi-replica shared rate limiting/state distribution requires deployment-specific infrastructure.
- Batch tools such as HF provenance scanning and adversarial robustness are intentionally not forced behind synchronous HTTP routes.

SCOPED EVIDENCE

- Cited Python 3.12 CI: 82 passed; 95.96% statement coverage, same test job.
- Source implements an authenticated gateway and four synchronous service slots. Local Compose exercises contract stubs.
- Cited CI includes successful stub health and integration jobs; production product images must be supplied externally.
- Image signing, attestation and release steps were skipped in the cited run. No deployed service, uptime or load SLO is established.

REPRODUCTION

- git checkout d54336b5f5f058a4d756bf2c7a023ad8510bbd6f
- Install the documented development dependencies; run python -m pytest tests/ -q. Use the cited job configuration when reproducing coverage.
- Benchmark data, adapters and environment requirements are separate; consult poster/13_reproducibility.md and the claim ledger.

Dated tests and committed fixtures do not establish universal detection, production deployment or an SLO.